Deep Research with Langgraph

Ravi Kumar Yadav, Nidhin Pattaniyil

About Us

Ravi Kumar Yadav
[Linkedin]
ravi@sukuya.com

Nidhin Pattaniyil
[Linkedin]
npatta01@gmail.com

ML Engineers @ Search

Resources

Github

https://github.com/npatta01/simple-deep-research

Slides

In repo (assets/pydata_2025.pdf)

Setup: CodeSpaces

Setup Validation: notebooks/00_setup.ipynb

Target Audience

- Beginner in building Agents / Agentic Workflow
- Learn the basics of using LangGraph
- Learn how to build your own deep research

Learning Objective

- What are the components of a Deep Research System
- How to use LLMs to scope a user’s problem
- How to use LLMs to perform tool use to search
- How to use LLMS to output a research brief
- How to orchestrate this as a workflow using LangGraph

Agenda

Deep Research in the Wild

Deep Research Service Providers

OpenAI

Gemini

Grok

perplexity

OpenAI Deep Research

what is the best coffee shop in Seattle

Could you share what matters most to you in a coffee shop? For example:

1. Vibe: Cozy, artsy, quiet, lively, etc.
2. Coffee Style: Third-wave espresso, pour-over, cold brew, etc.
3. Food Options: Pastries, full brunch, vegan items?
4. Location: Any neighborhood in Seattle you prefer?
5. Seating/Work Needs: Good for working or just grabbing a quick cup?

This'll help me pinpoint the best fit for you.

OpenAI Deep Research

ambiance, good for working, pastries

Got it! I'll look for the best coffee shops in Seattle that have great ambiance, are ideal for working, and offer delicious pastries.

I'll let you know as soon as I've got a shortlist for you. Hang tight!

Share ···

Activity 19 Sources

ambiance, good for working, pastries

Got it! I'll look for the best coffee shops in Seattle that have great ambiance, are ideal for working, and offer delicious pastries.

I'll let you know as soon as I've got a shortlist for you. Hang tight!

Research completed in 14m · 19 sources · 63 searches

Best Seattle Coffee Shops for Ambiance, Work & Pastries

Seattle's café scene offers more than just great coffee – many shops combine inviting atmospheres, work-friendly settings, and delicious baked goods. Below is a ranked shortlist of top Seattle coffee shops excelling in cozy or stylish **ambiance**, remote-work **suitability** (ample seating, Wi-Fi, reasonable noise), and **quality pastries** (fresh, locally sourced, artisan treats). Each entry notes location (neighborhood), notable hours or features, and standout offerings.

1. Temple Pastries (Central District)

Ask anything

Best Seattle Coffee Shops for Working with Great Ambiance and Pastries

I'm considering a range of Seattle coffee shops, focusing on great ambiance, suitability for working, and quality pastries. Notable mentions include "Ada's Technical Books & Cafe" and "Third Place Books & Cafe".

Searched for coffee shops in Seattle with pastries and wifi

Consulting Yelp, Reddit, Seattle Eater, Zach Taiji site, Seattle Refined, Infatuation, local publications, and Seattle coffee shop institutions for detailed narratives.

Searched for best Seattle coffee shops with pastries

I'm reviewing lists from Forbes, Seattle Eater, Reddit, and various blogs to find Seattle coffee shops with good ambiance, wifi, and quality pastries for working.

Read seattle.eater.com

OK, let's keep working on the Eater listing. The goal is clear—ensuring comprehensive coverage and clarity in the listing.

Read more from seattle.eater.com

what is the best coffee shop in Seattle

Here's the plan I've put together. Let me know if you need any changes before I start researching.

Seattle's Best Coffee Shops

Research Websites

(1) Identify recent articles, rankings, and 'best of' lists from reputable sources concerning top coffee shops in Seattle, Washington.
(2) Search for coffee shops in Seattle that have won regional or national awards for their beans, brewing quality, or espresso.
(3) Investigate local coffee roasters based in Seattle and check if they operate retail locations frequently cited for exceptional quality....

More

Analyze Results

Create Report

Ready in a few mins

Edit plan | Start research

Gemini

ChatGPT

Intro (Brief)

Place 1:
- Why it stands out
- Hours

Place 6:
- Why it stands out
- Ambiance
- Pastries
- Hours

14 min;
19 sources;
63 searches

Link, Link (Personal)

Gemini

Executive Summary

Evaluation Methodology

Deep Dive Place 1-3:
- Ambiance
- Productivity
- Culinary

Pastry Benchmarks

Comparison

Conclusions

5 min;
24 sources;
?? searches

Link, Link(Personal)

Key Transitions

- **Identifying Key Requirements**
- Multi-Criteria Approach
- **Starting Research**
- *Researching websites*
- Identifying Top Contenders
- Resolving **Information Gaps**
- **Deep Dive Strategy**
- *Researching websites*
- Strong Shortlist Developed
- **New Insights on Pastry and Work Suitability**
- Refining the Final Comparison
- *Researching websites*
- **Comprehensive Synthesis Complete**
- Defining Top Models
- Finalizing Nuanced Recommendations
- **Preparing Final Report**

LLM Basics

Notebook

Notebook: 01_llm_basics.ipynb

Sections:

- LLM and Memory
- Structured Outputs
- Tool Calls

Structured Output

```python
class AnimalInfo(BaseModel):
    name:str
    first_discovered:str
    latin_name: str
    num_species:str
```

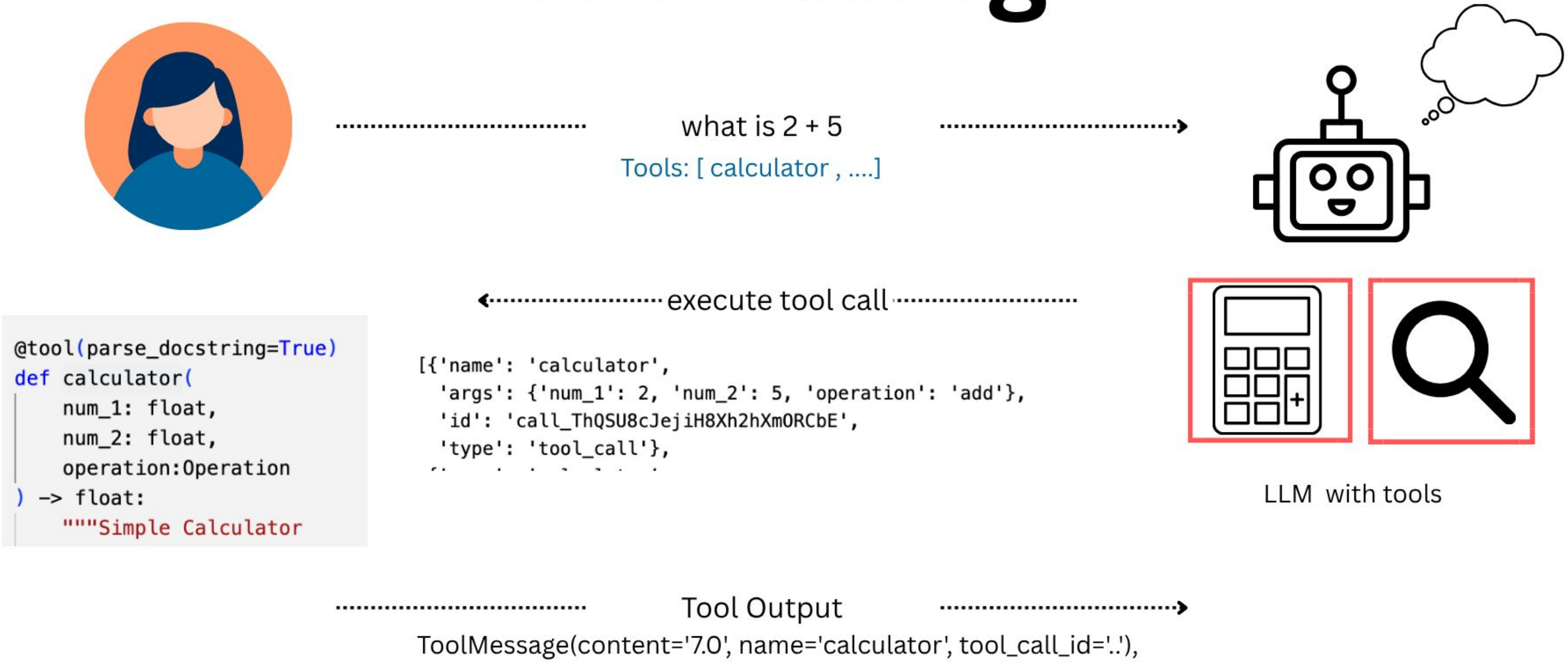
```python
model.with_structured_output(AnimalInfo)
```

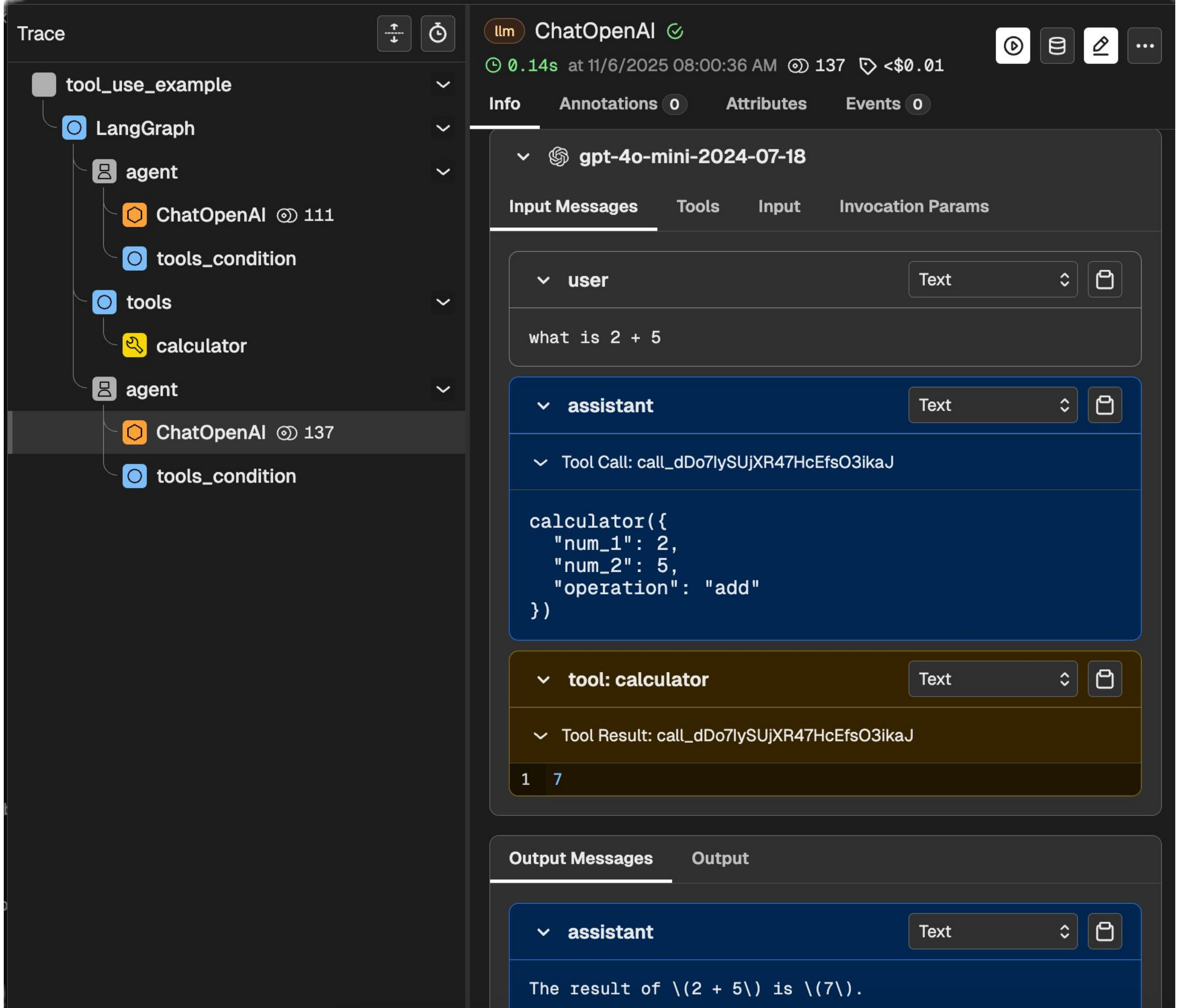
```
AnimalInfo(
    name='Elephants',
    first_discovered='Paleolithic period',
    latin_name='Loxodonta (African elephants) & Elephas (Asian elephants)',
    num_species='Three species: African savanna, African forest, and Asian elephant.'
)
```

LLM Basics

- Tool
  - Python functions
  - Enables interactions
    - APIs
    - Databases
    - File Systems

Tool Calling




Trace
tool_use_example
LangGraph
agent
ChatOpenAI 111
tools_condition
tools
calculator
agent
ChatOpenAI 137
tools_condition
llm ChatOpenAI
0.14s at 11/6/2025 08:00:36 AM 137 <$0.01
Info
Annotations 0
Attributes
Events 0
gpt-4o-mini-2024-07-18
Input Messages
Tools
Input
Invocation Params
user
Text
what is 2 + 5
assistant
Text
Tool Call: call_dDo7lySUjXR47HcEfsO3ikaJ
calculator({
"num_1": 2,
"num_2": 5,
"operation": "add"
})
tool: calculator
Text
Tool Result: call_dDo7lySUjXR47HcEfsO3ikaJ
1 7
Output Messages
Output
assistant
Text
The result of \(2 + 5\) is \(7\).

Langgraph Basics

Notebook

Notebook: 01_langraph_basics.ipynb

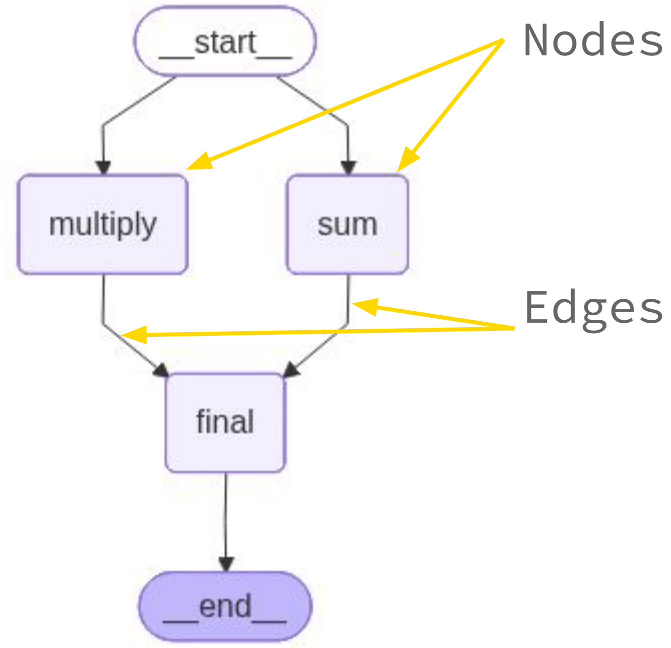
Sections:

- Define Graphs
- Define Deep Research Graph

Graph Basics

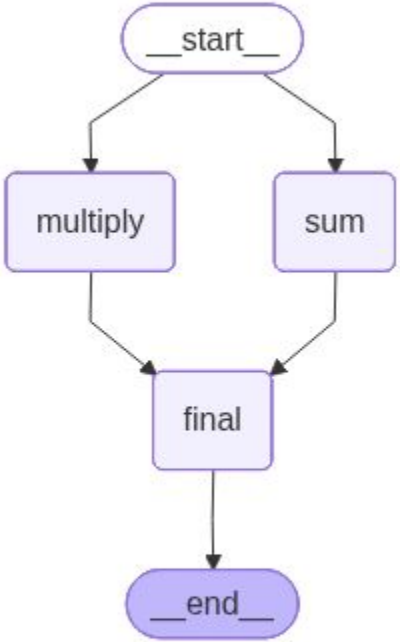
- Nodes
  - Python Functions (LLMs/Tools)

- Edges
  - Connector between two nodes
  - Types
    - Normal
    - Conditional



Graph Basics

- State
  - Python Object containing data
  - State is transmitted over Edges between Nodes.

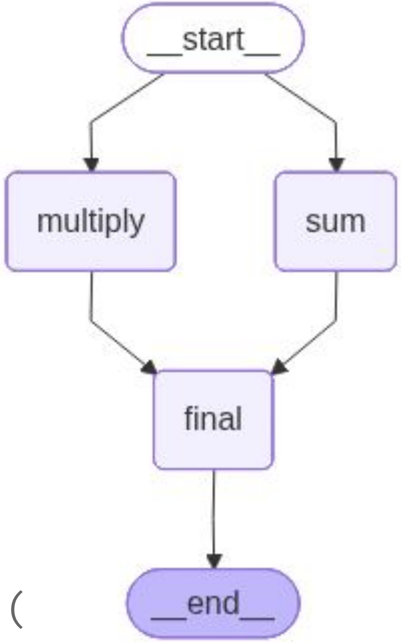


Simple Graph

```python
class MathState(BaseModel):
    a: int
    b: int
    total_sum: int | None = None
    total_multiply: int | None = None
    total_final: int | None = None


def compute_sum(state: MathState) -> dict[str, Any]:
    return {"total_sum": state.a + state.b}

def compute_product(state: MathState) -> dict[str, Any]:
    return {"total_multiply": state.a * state.b}

def compute_final(state: MathState) -> dict[str, Any]:
    return {"total_final": state.total_sum + state.total_multiply}
```

```python
single_node_graph = StateGraph(MathState)
single_node_graph.add_node("sum", compute_sum)
single_node_graph.add_node("multiply", compute_product)
single_node_graph.add_node("final", compute_final)
single_node_graph.add_edge(START, "sum")
single_node_graph.add_edge(START, "multiply")

single_node_graph.add_edge("sum", "final")
single_node_graph.add_edge("multiply", "final")
single_node_graph.add_edge("final", END)

single_node_app = single_node_graph.compile()
```

```
single_node_app.invoke(
  {“a”:5, “b”: “2”}
)
```

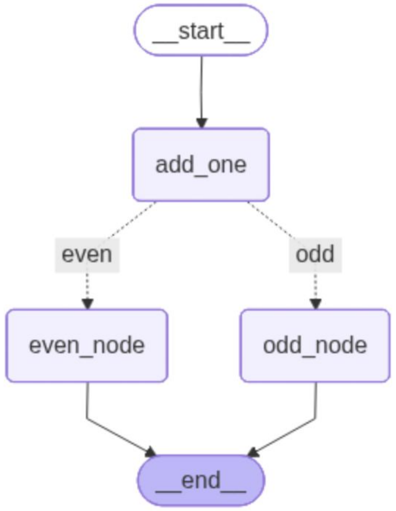


Conditional Graph

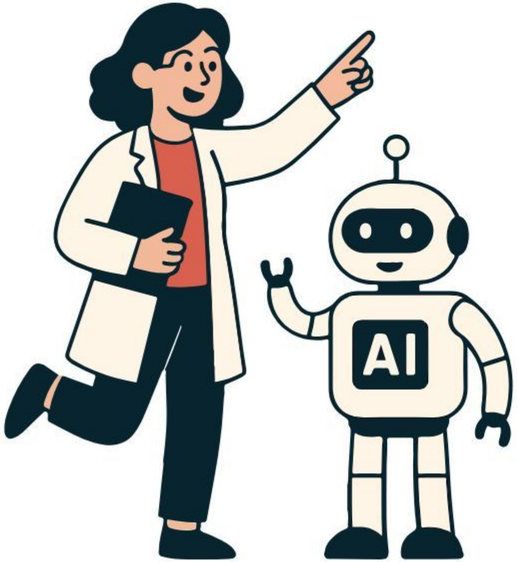
```python
class State(BaseModel):
    number: int
    final_message: str = None
```

```python
def add_one_node(state:State) -> State:
    return {"number": state.number + 1}

def even_node(state:State) -> State:
    return {"final_message":"number is even"}

def odd_node(state:State) -> State:
    return {"final_message":"number is odd"}

def number_checker_condition(state:State) ->Literal["even","odd"]:
    if state.number % 2 == 0:
        return "even"
    else:
        return "odd"
```

```python
workflow.add_edge(START, "add_one")
workflow.add_edge("odd_node",END)
workflow.add_edge("even_node",END)

# Add conditional edges depending on number
workflow.add_conditional_edges(
    "add_one",
    number_checker_condition,
    {"even": "even_node", "odd": "odd_node"}
)
```

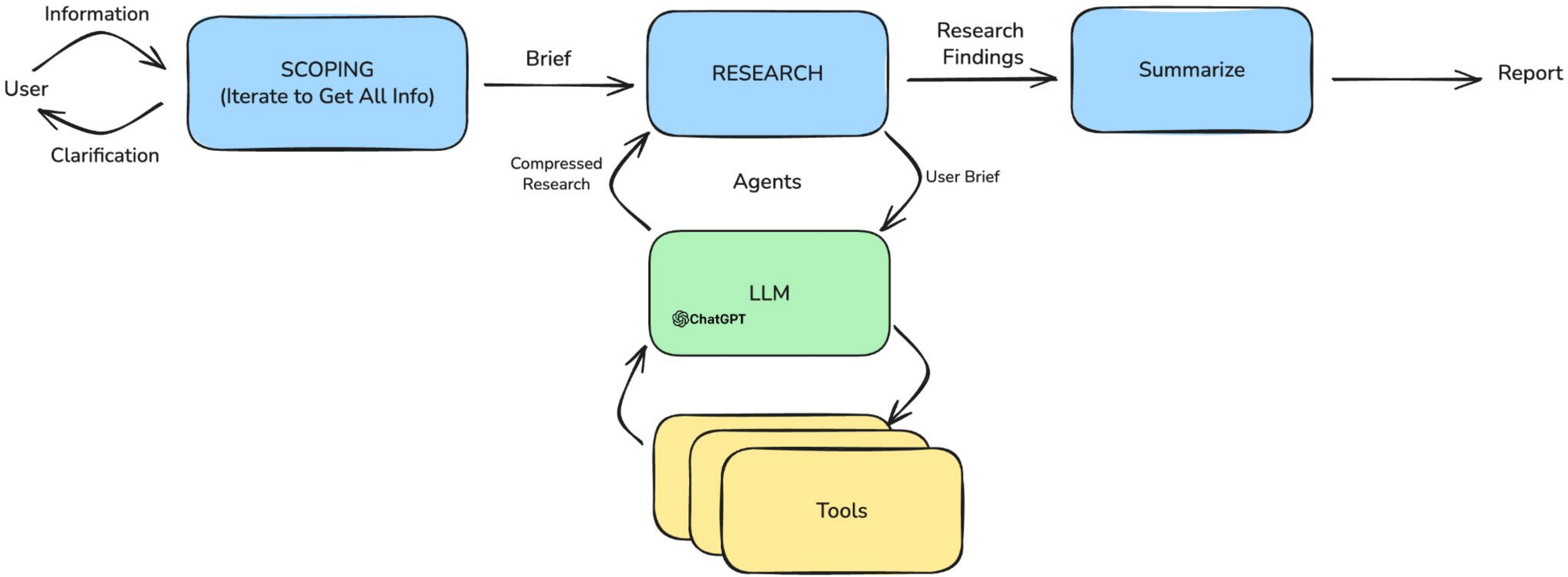


LangGraph



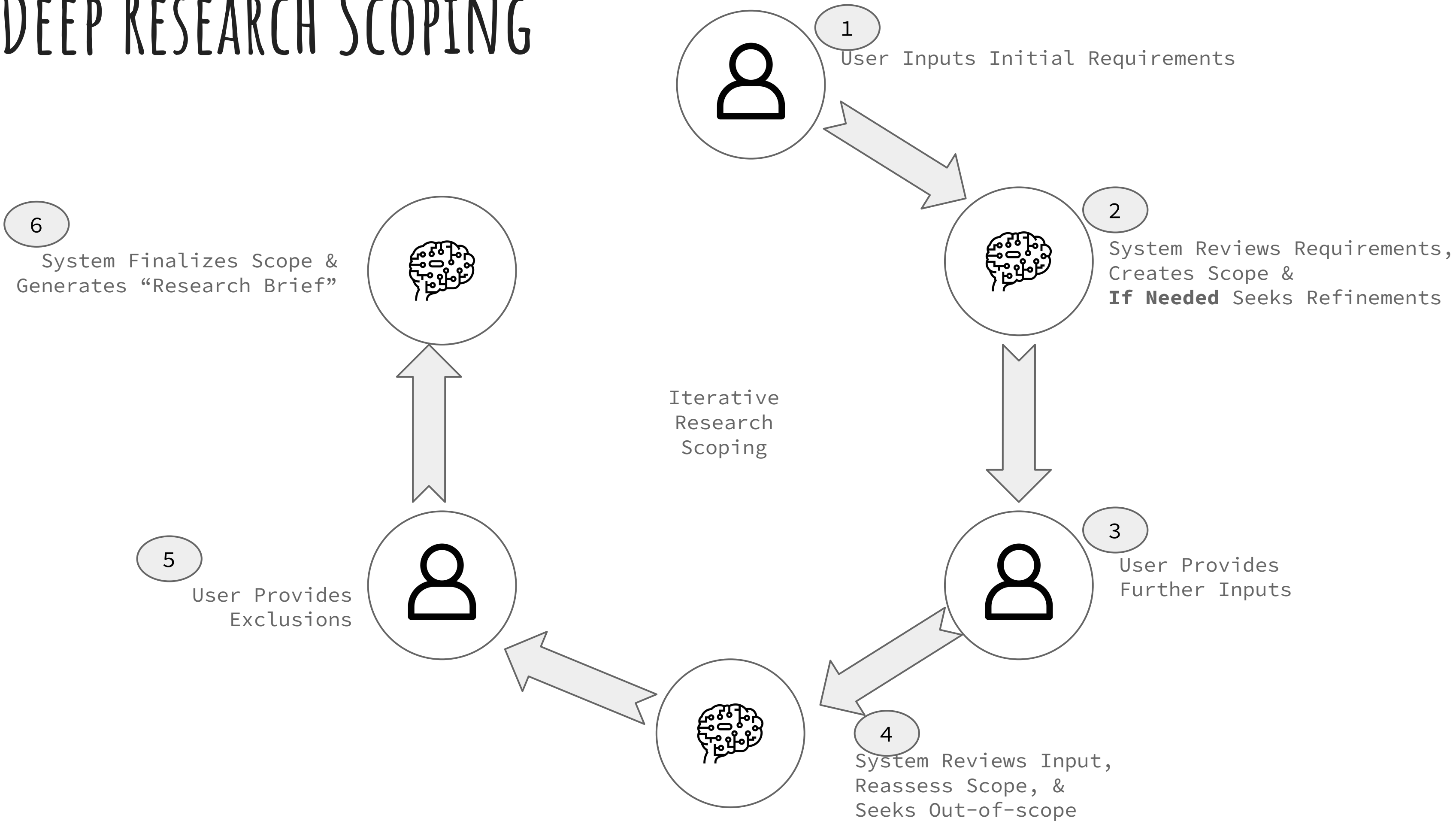
Deep Research Components

High-level Deep Research system overview



Scoping a problem

Deep Research Scoping



Notebook

Notebook: 02_scoping.ipynb

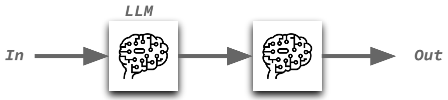
Section:

- Iteratively interact with the user to clearly define the research goals.
- Generating research brief

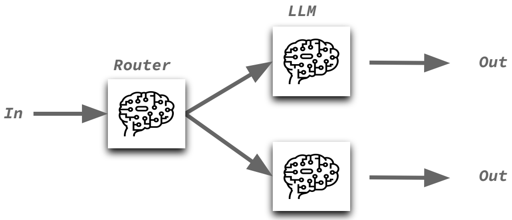
Workflow and Agents

Design Patterns - "deterministic"

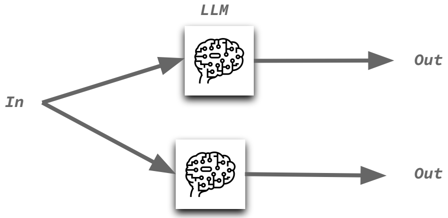
Prompt Chaining



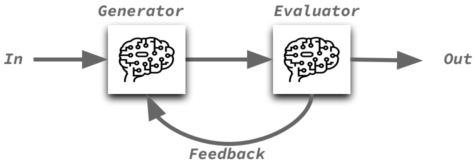
Routing



Parallelization

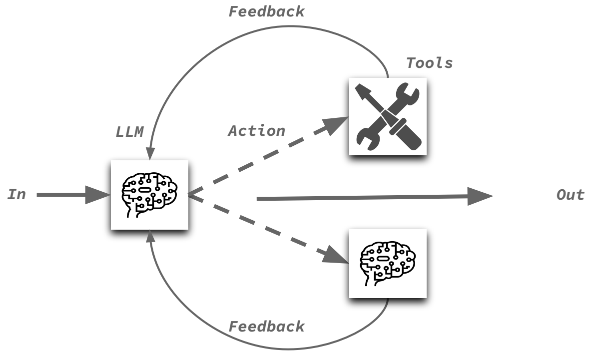


Evaluator-Optimizer



<u>Workflow Pattern</u>:Fixed deterministic path

Design Patterns - "Non-deterministic"

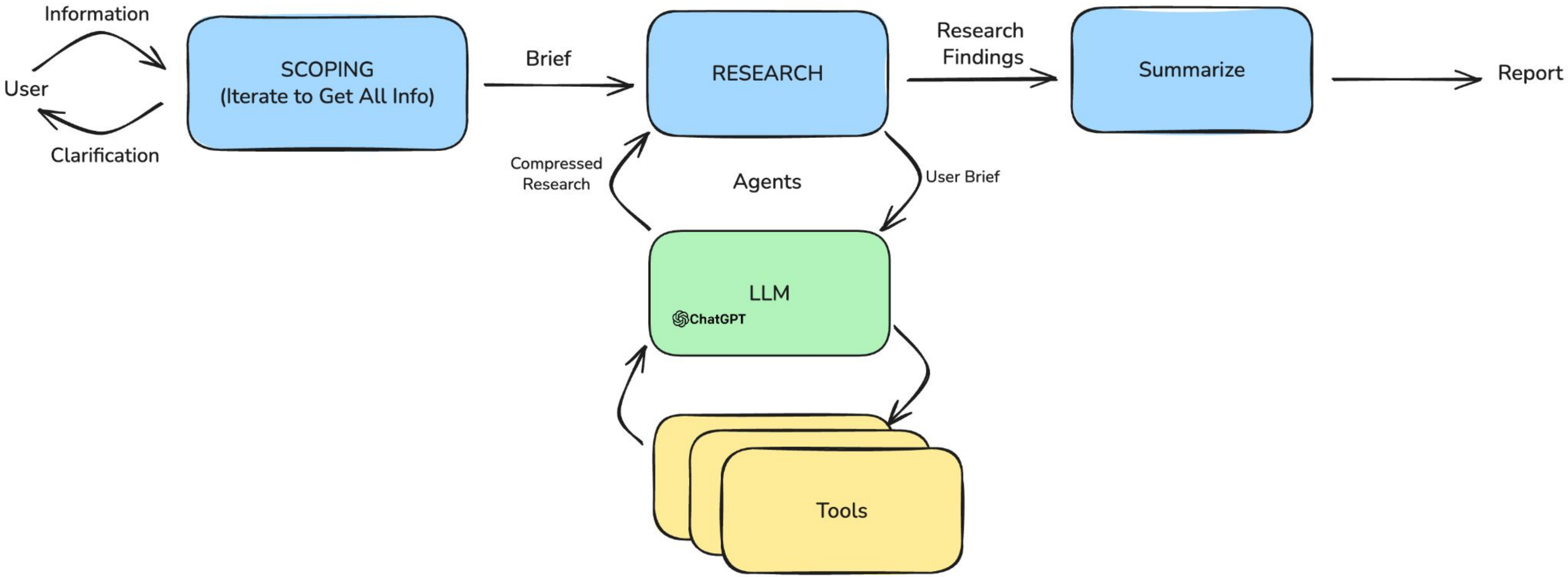


<u>Agent Pattern</u>:

- Autonomous and non-deterministic path
- Autonomy to pick any available tool

Building effective agents from Anthropic: link

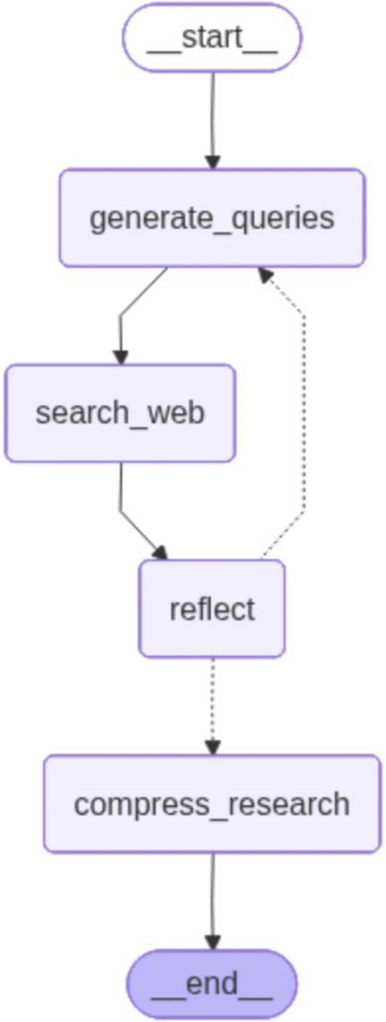
High-level system overview



Researcher Workflow

Core Steps in ReSearch

- Generate Queries
- Search the web with the queries
- Reflect on the information extracted so far
- Repeat the information as needed
- Synthesize / Compress the information



Notebook

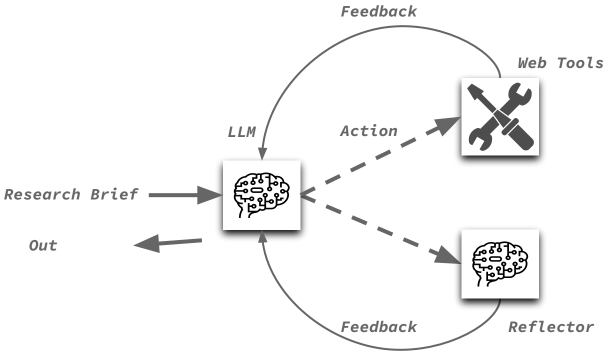
Notebook: 03_research_as_workflow.ipynb

Section:

- Research step as a workflow

Researcher Agent using Tools

Researcher Agent



Input: Research Brief
Actions:

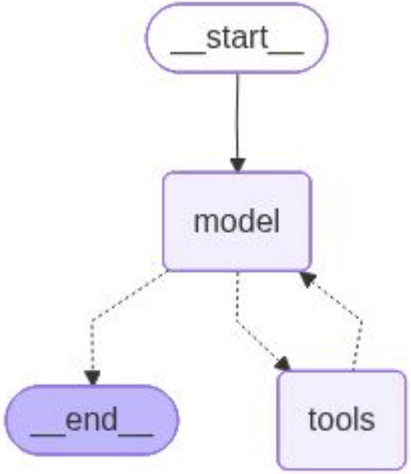
- Web Tools
- Reflector
- Compressor

Output: Formatted Compressed Summary

Defining an Agent

```python
agent = create_agent(
    llm,                            # The LLM to use
    tools,                          # List of tools
    system_prompt=system_message # System prompt
)
```

```python
@tool
def web_search_tool(query: str, max_results: int = 3) -> str:
    """Search the web for information about a specific query.

    Args:
        query: The search query (be specific for better results)
        max_results: Number of results to return (1-5)

    Returns:
        Formatted search results with sources
    """

    results = utils.web_search(query, max_results=max_results)

    return results
```



Notebook

Notebook: 03_research_as_agent.ipynb

Section:

- Research step as an agent

Report Generation

Notebook

Notebook: 04_write_report.ipynb

Section:

- Taking research notes and write report
-

REPORT

The Best Coffee Shops in Seattle (2025): Focus on Coffee Bean Quality and Atmosphere

Seattle, renowned for its rich coffee culture and as the birthplace of several coffee innovations, boasts a diverse selection of coffee shops ranging from cozy local spots to immersive retail experiences. This report identifies leading coffee shops within the city by prioritizing the quality of their coffee beans and the atmosphere of their establishments. The evaluation draws on multiple recent guides and reviews, ensuring balanced coverage across firsthand experiences and expert curation.

Criteria for Selection

Two primary criteria informed this assessment:

- **Quality of Coffee Beans:** Shops and roasters known for exceptional sourcing, flavor, and ethical practices.
- **Atmosphere:** Environments that enhance the coffee experience, whether through design, ambiance, or service.

Top Contenders: Seattle’s Exemplary Coffee Shops

Storyville Coffee Company

Located in Pike Place Market, Storyville Coffee Company consistently ranks among Seattle's finest coffee destinations. It garners a notable 4.5/5 rating from over 2,500 reviewers, signaling both enduring popularity and high customer satisfaction. The shop is described as providing "expensive coffee vibes," reflecting not only premium pricing but also a luxurious ambiance and attention to detail in its offerings. The central location makes it a popular stop for tourists wishing to experience Seattle's coffee tradition in a bustling, historic setting. While prices may be higher relative to some competitors, the ambiance and quality are considered commensurate by many patrons TOP 10 BEST Coffee Roasters in Seattle, WA – Updated 2025 – Yelp, STORYVILLE COFFEE COMPANY – Updated November 2025 – Yelp.

Starbucks Reserve Roastery

Central Seattle is home to the flagship Starbucks Reserve Roastery, which offers an immersive, upscale coffee experience beyond the standard café format. It stands out for its rare brews, on-site roasting demonstrations, and comprehensive service—features that attract both casual visitors and coffee aficionados. Service levels are routinely described as "exceptional," with knowledgeable and attentive staff who contribute significantly to the shop's reputation. Additionally, the Roastery functions as a significant tourist attraction, enhancing the overall ambiance and making it a destination in itself rather than just a place to get coffee Seattle Coffee Guide 2025: The Best Cafes for Work, Tourists ..., Starbucks Reserve Roastery Seattle (2025) – Tripadvisor.

Olympia Coffee

Olympia Coffee is frequently noted for its combination of direct-trade, ethically sourced beans and a thoughtfully designed café environment. The space is described as comfortable and work-friendly, with a clean, modern aesthetic and ample seating. Coffee enthusiasts highlight the shop’s focus on quality and sustainability, citing the direct relationships with coffee growers that underpin the establishment’s mission. The result is a range of coffee options that appeal to both connoisseurs and those seeking a welcoming place to work or socialize Seattle Coffee Guide 2025: The Best Cafes for Work, Tourists ....

Onda Origins

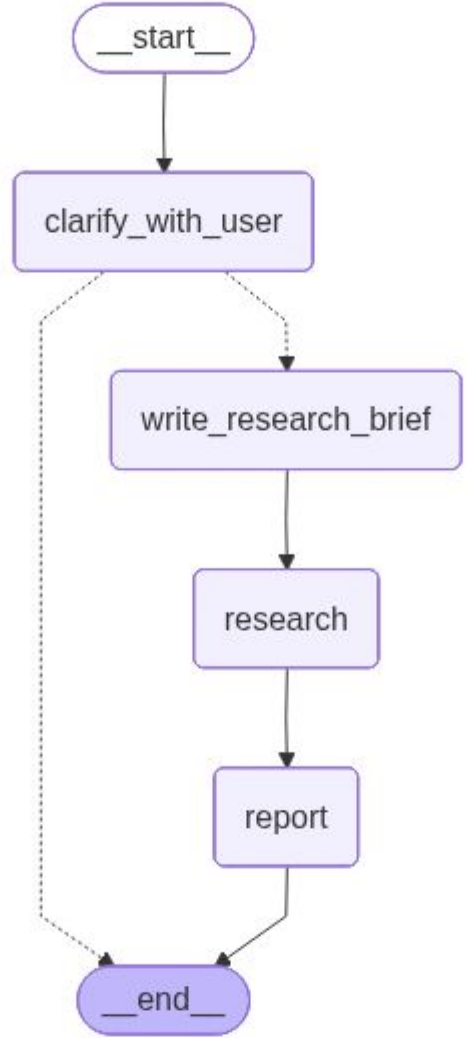
Final Graph

Notebook

Notebook: 05_full_graph.ipynb

Section:

- Full E2E graph
-



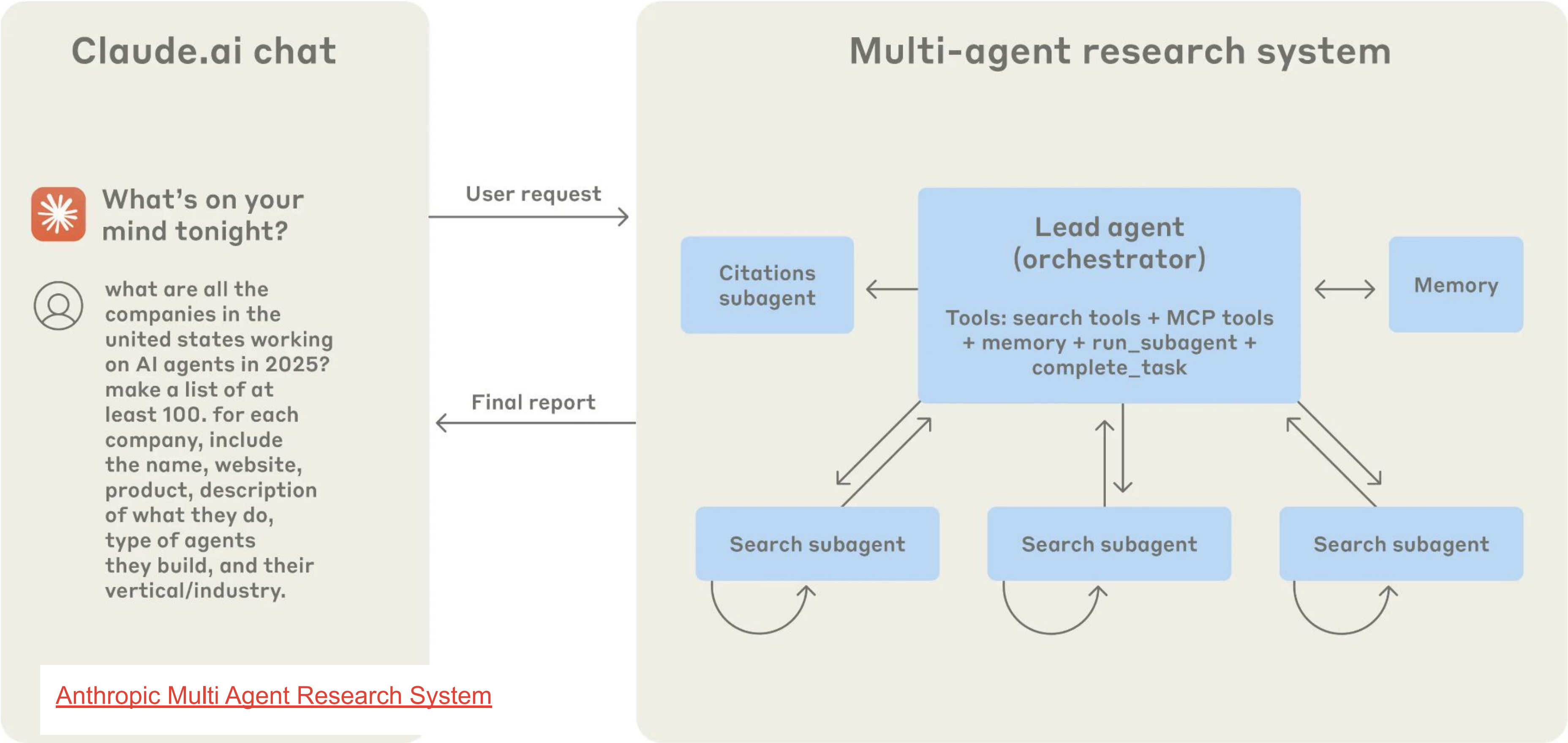
Self hosting options

Open Self hosted options

- LangChain Open Deep Researcher
- Salesforce Enterprise Deep Researcher
- Nvidia Deep Researcher

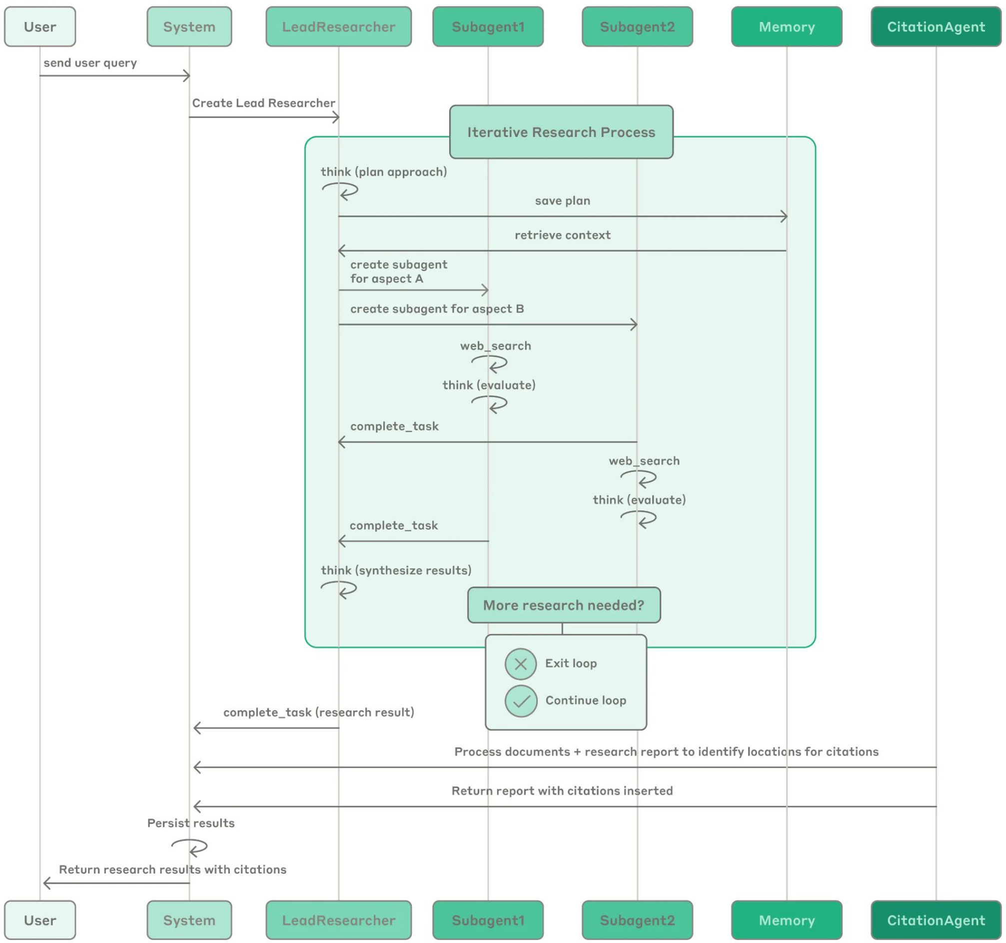
Industry system

High-level Architecture of Advanced Research



Anthropic Multi Agent Research System

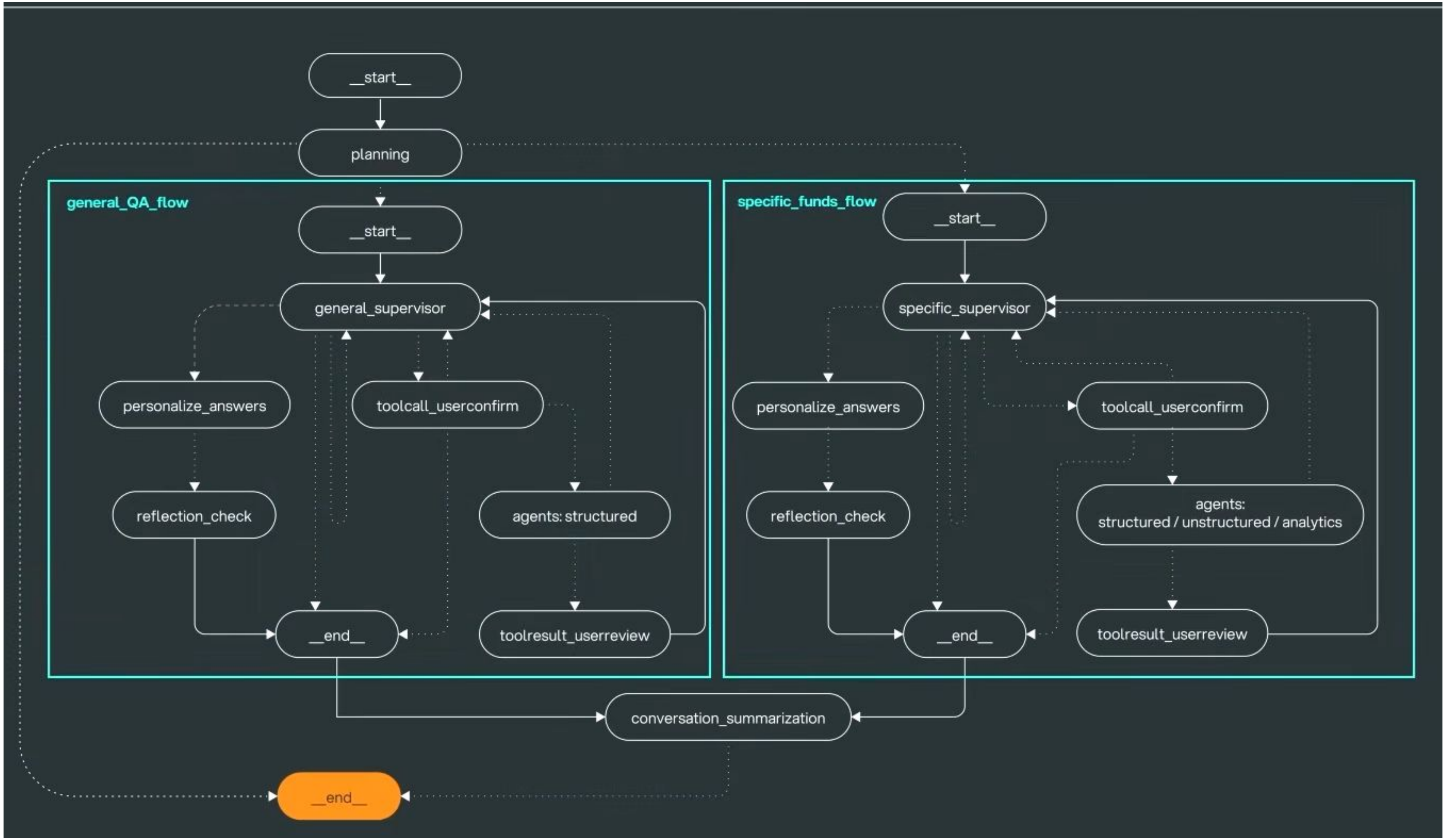
Multi-agent System Process Diagram



JPMorgan: Data Analytics, Visualization, Insights, and Decision-making Assistant

- Graph with multiple subgraphs
- Subgraphs have can have 4 different agents:
  - Supervisor Agent
  - Structured Data Agent
  - Unstructured Data Agent
  - Analytics Agent

How JP Morgan Built An AI Agent for Investment Research with LangGraph | LangChain Interrupt



__start__
planning
general_QA_flow
__start__
general_supervisor
personalize_answers
toolcall_userconfirm
reflection_check
agents: structured
__end__
toolresult_userreview
specific_funds_flow
__start__
specific_supervisor
personalize_answers
toolcall_userconfirm
reflection_check
agents:
structured / unstructured / analytics
__end__
toolresult_userreview
conversation_summarization
__end__

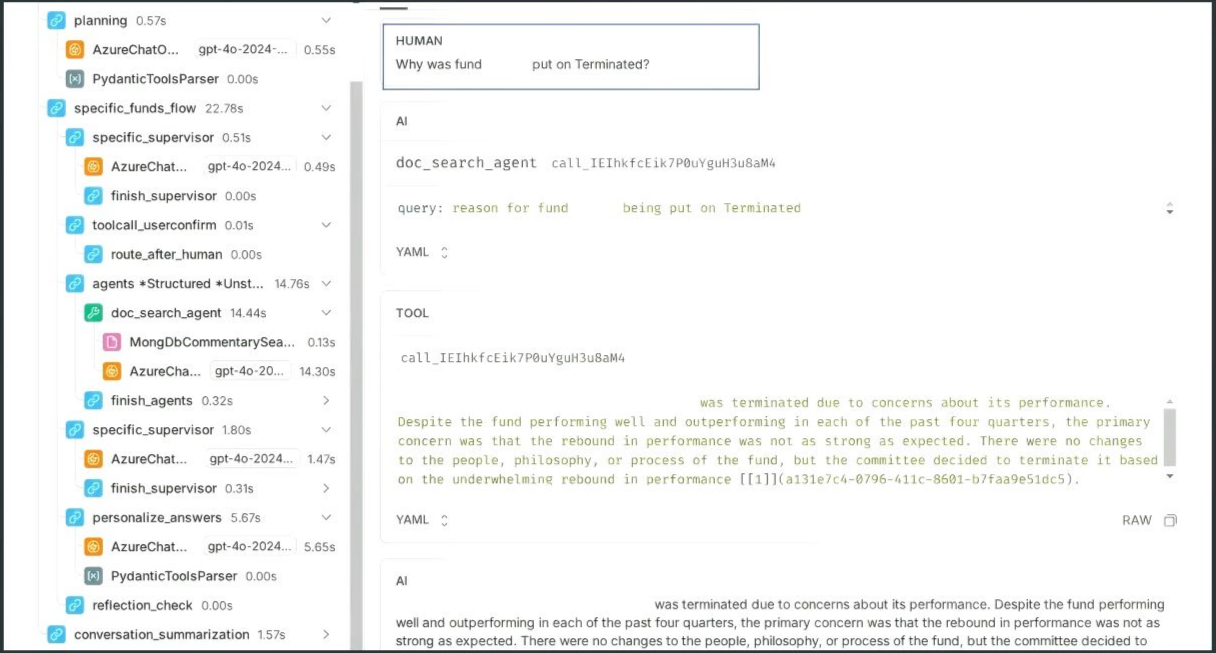
Example: Why was a fund terminated?

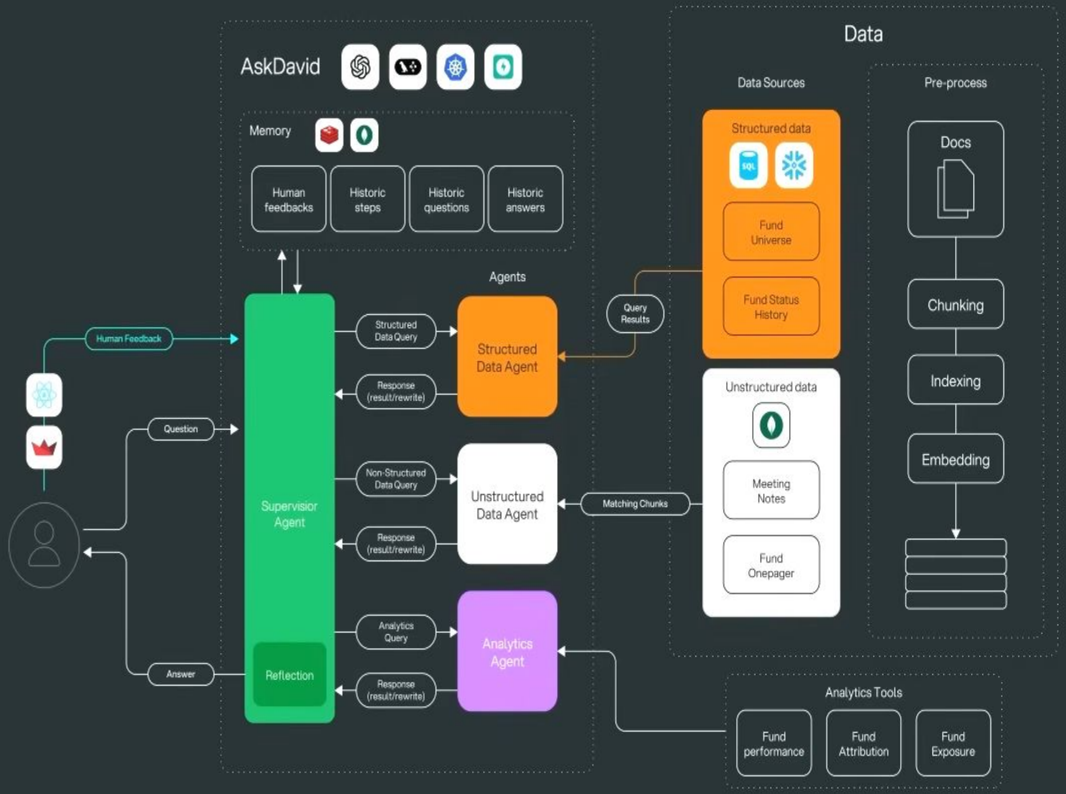
Planning
- Determines the intention
- Invokes the corresponding subgraph

Subgraph - Specific Funds Flow
- delegates to doc_search_agent

Doc Search Agent
- Conducts mongoDB search to retrieve relevant information

Supervisor
- Personalization based on the role of human
- Reflect to ensure the accuracy and relevancy of answers
- Summarize conversation, update memory, and return answer





Supervisor Agent:

- Acts as the orchestrator
- It receives questions from users and determines a plan to provide answers
- This may involve engaging one or more of the specialized sub-agents, depending on the nature of the query
- It incorporates human in the loop to ensure the accuracy and reliability
- It also has access to long term memory

Structured Data AI Agent:
This agent translates natural language queries into SQL or GraphQL to retrieve answers from databases or APIs, and then uses a large language model (LLM) to summarize the results.

Unstructured Data AI Agent:
This RAG agent utilizes a vector database and hierarchical index retrieval to store documents and search for relevant information chunks, followed by summarization using an LLM.

Analytics Agent:
ReAct agent with access to analytics APIs as tools.

Evaluation

Deep Research Bench

- Evaluate DeepResearch implementations on 100 PHD level related tasks in 22 different domains
- Evaluated
  - Comprehensiveness: Coverage of relevant information
  - Insight/Depth: Quality of analysis and reasoning
  - Instruction-Following: Adherence to task requirements
  - Readability: Clarity and organization of the report
  - Citation Accuracy: Percentage of correctly supported citations
  - Effective Citations: Average number of verifiably supported statements per task

DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents

Deep Research Leaderboard

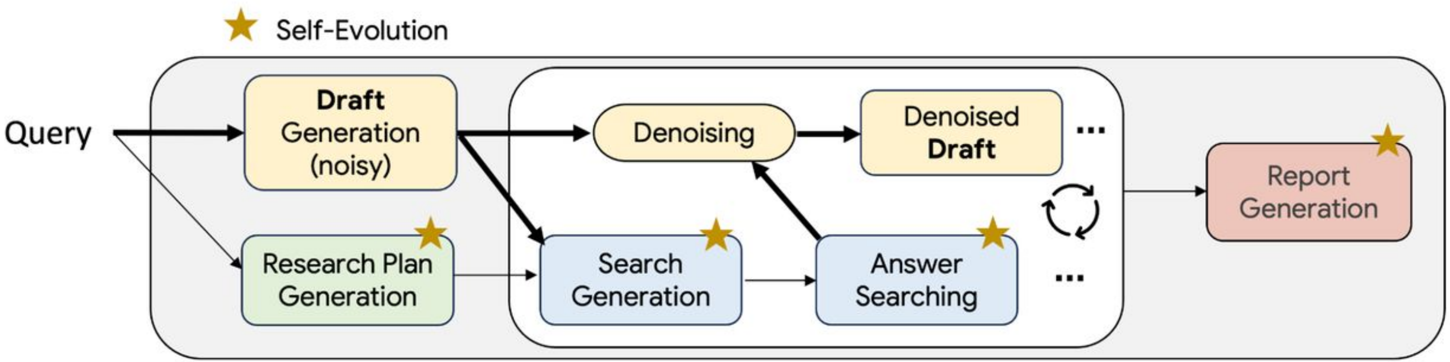
Rank	model	overall	comprehensiveness	instruction following	Readability
1	gemini-2.5-pro-deepresearch	**49.71**	**49.51**	**50.12**	**50**
2	openai-deepresearch	46.45	46.46	49.39	47.22
3	claude-research	45	43.54	47.58	44.66
6	langchain-open-deep-research	43.44	42.97	48.09	45.22
7	nvidia-aiq-research-assistant	40.52	37.98	44.59	42.63
8	perplexity-Research	40.46	39.1	46.11	43.08

overall: Overall Score (weighted average of all metrics)
comp.: Comprehensiveness - How thorough and complete the research is
insight: Insight Quality - Depth and value of analysis
inst.: Instruction Following - Adherence to user instructions
read.: Readability - Clarity and organization of content

https://huggingface.co/spaces/Ayanami0730/DeepResearch-Leaderboard

Future iterations

Deep Researcher with Test-Time Diffusion



models research report writing as a diffusion process, where a messy first draft is gradually polished into a high-quality final version

https://research.google/blog/deep-researcher-with-test-time-diffusion/

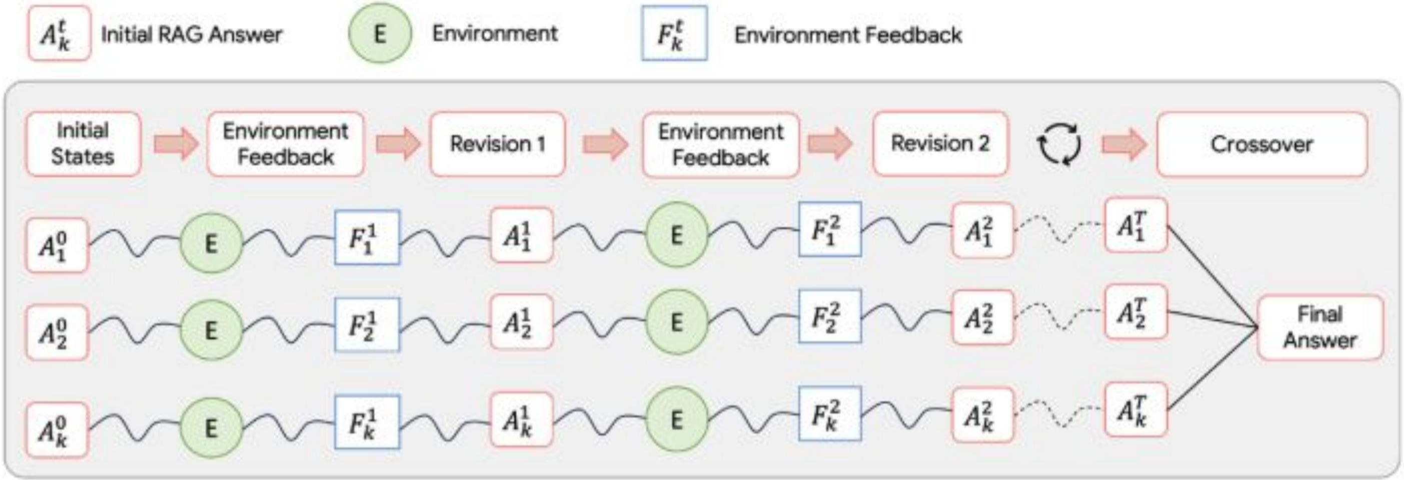


Illustration of the component-wise self-evolution algorithm applied to Search Answer (Stage 2b). The process starts with multiple variants of initial answers, each undergoing a self-evolving episode where it first interacts with the environment to obtain a fitness score and feedback. It is then revised based on the feedback. This process repeats until the maximum number of iterations is reached. Finally, multiple revised variants from all episodes are merged to produce the final answer.

Deep Research with Vision Support

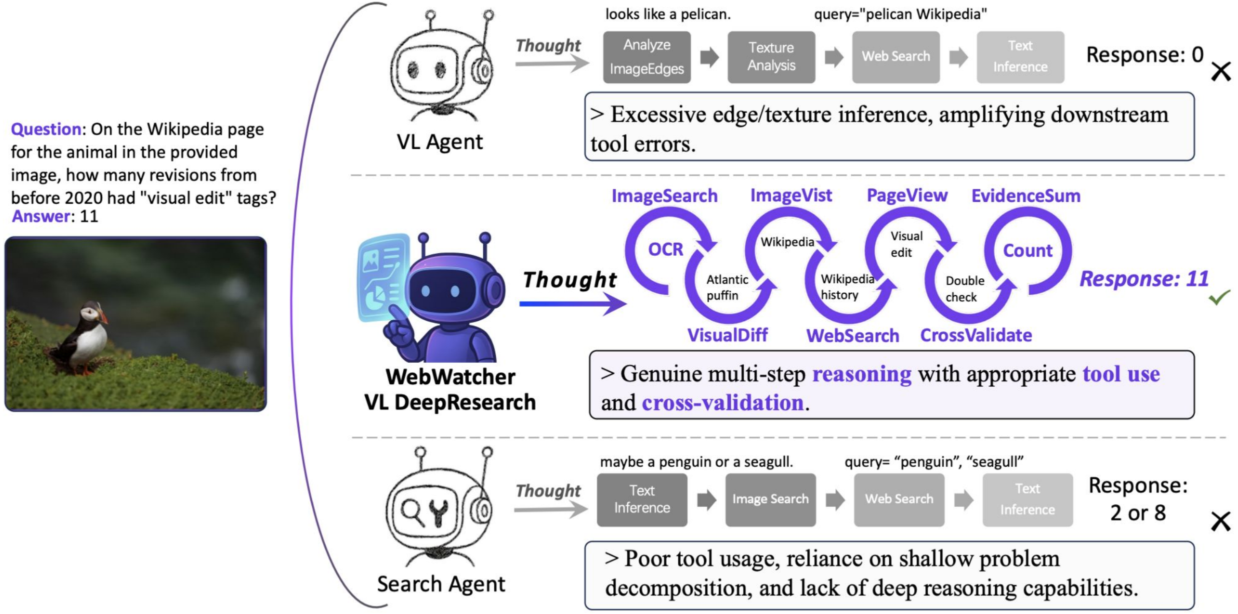


Figure 2: Comparison of VL reasoning agents. WebWatcher resolves the GAIA case that defeats either vision-only reasoning or search-based agents, demonstrating the strength of multi-tool integration and in-depth reasoning generalization.

Tongyi Lab , Alibaba Group

https://tongyi-agent.github.io/blog/introducing-tongyi-deep-research/

Questions

References

- Langraph Deep Research Course
- Anthropic: How we built our multi-agent research system
- Deep Research Agents: A Systematic Examination And Roadmap
- DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents
- How Deep Research Works - Mukund Sridhar & Aarush Selvan, Google DeepMind
- Open Implementation: LangChain, Salesforce, Nvidia